

Human Factors and Ergonomics

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Definition

In its triennial report, the International Ergonomics Association (IEA, 2000) defined ergonomics as the scientific discipline that deals with understanding the interaction between humans and other elements of a sociotechnical system. In this definition, ergonomics is the profession that applies theory, principles, data, and design methods to optimize human well-being and the overall performance of a system. It is in particular responsible for the design and evaluation of tasks, jobs, products, environments, and systems to make them compatible with the abilities, needs, and limitations of people.

The word “ergonomics” comes from the Greek “ergon” meaning work and “nomos,” which means law. Therefore, etymologically, this is the science of work. The term has been used historically in the European tradition. In the American tradition the term “human factor engineering” is used to refer to the same issues, so both terms can be now considered as synonyms and are used interchangeably. The latter is evidenced by the fact that the “Human Factor Society,” founded in Tulsa (Oklahoma) in 1957, is now called the “Human Factor and Ergonomics Society” (HFES). Another term which is often used in the same context is “Engineering psychology” (Wickens & Hollands, 2000).

The early precursors of the new discipline could be set around the time of World War I. They had their background in the pioneering studies of Frederic Bartlett (1886–1969), Hugo Münsterberg (1863–1916) and Frederick Winslow Taylor (1856–1915) on applied psychology and industrial management. The design of new machines (e.g. the first cars or tanks) revealed the importance of taking into account the characteristics of the people who should operate them. It was found that many

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people had difficulties in operating more complex machines, especially warplanes. This led the army to recruit psychologists, who were assigned the task of developing and administering tests to select soldiers and to assign them to different tasks. These applied psychologists set up the first human factors laboratories, which continued their work after the war ended. But it was World War II that provided the final impetus for the establishment of ergonomics as a discipline with industrial and academic recognition. Moreover, this war involved an enormous amount of people and artifacts, many of them newly created, such as radar, which made the idea of selecting a few special individuals to use previously designed artifacts unworkable. The idea that emerged, and has had an enormous impact on the development of the discipline, was that the devices should be designed taking into account the characteristics of the human beings who will use them, and not adapted to people once they are designed.

In Europe, the focus of ergonomics is to be found in industry, and it has been linked to an interest in improving worker performance and satisfaction. The discipline began with an emphasis on the design of equipment and workplaces, although, in principle, it dealt with biological, rather than psychological aspects. In this way, studies began on anthropometry, work medicine, architecture, lighting, and so on. Back in the 1980s, European ergonomists began to worry largely about advanced psychological aspects and the “European Association of Cognitive Ergonomics” (EACE) emerged, leading to a confluence of interests with human factors and cognitive science professionals on the other side of the Atlantic.

The definition of ergonomics is extended today to all human activities in which artifacts are utilized. Ergonomists (with many applied psychologists among them) are involved in a permanent search for comprehensive approaches in which the physical, cognitive, social, and environmental aspects of human activities can be considered. Although ergonomists often work on different economic sectors or particular tasks, these application domains are constantly evolving, creating new ones and changing the perspective of the old ones. Accordingly, one can recognize today four main domains of expertise crucial for investigating interaction between humans and socio-technical systems.

- *Physical ergonomics* deals with the anatomical, anthropometric, physiological, and biomechanical parameters in static and dynamic physical work. Among the main topics are the physical postures that people adopt when they are working, fatigue, and other problems associated with handling physical and musculoskeletal tasks associated with physical efforts.
- *Cognitive ergonomics* is a subdiscipline of ergonomics that studies the cognitive processes at work with an emphasis on an understanding of the situation and on supporting reliable, effective, and satisfactory performance. This approach addresses problems such as attention distribution, decision making, formation of learning skills, usability of human–computer systems, the cognitive aspects of mental load, stress, and human errors at work.
- *Neuroergonomics* is a relatively new development that involves the application of more in-depth neurophysiological methods such as brain imaging techniques. This advanced methodology can be used for evaluating the customers’ preferences

for one or another design of human–computer interface or for a particular version of industrial products (this latter task is sometimes related to the field of “neuromarketing”).

- *Social or organizational ergonomics* deals with the optimization of sociotechnical work systems, including their structures, policies, and organizational processes. Thus, ergonomists are often involved in the social design of communication systems, interaction routines within the working groups, times and shifts schedules in a company, and other related issues.

By its very definition ergonomics is a multidisciplinary endeavor. Scientists and practitioners from many disciplines have been interested in designing or changing different aspects of sociotechnical systems. Therefore, they all might be called “ergonomists” in a sense, with further specifications depending on particular emphasis—from the physical and neurophysiological to the mental and socio-organizational aspects—of the interest at hand.

Areas of Application

In this chapter, the primary emphasis will be on ergonomic issues related to applied psychology. Its interest in complex behaviors, mental activities, and the resources of human information processing puts cognitive ergonomics in the focus of present analysis. Even with such constraints, however, it is difficult to make a straightforward classification of domains of application because ergonomists work in all areas of life in which there is a cognitive design problem. A simple classification based on two criteria is elaborated below.

Industrial areas

Human–computer interaction. For many years, the computer has been the most sophisticated artifact in our lives, visibly present both within and outside the workplace. Due to the overwhelming importance of information technologies, the term cognitive ergonomics is often replaced by Human Computer Interaction (HCI or, from a more technical perspective, CHI). However, in most cases, computers are only parts serving the functioning of larger technical systems, so our interaction with them is not as explicit as when a personal computer is in use. For example, there are computing devices in the engines of modern cars but drivers interact with the car and not with these “invisible computers.” For this reason, one should talk rather about the cognitive ergonomics of human-machine interaction and rethink interaction with computers as interaction with everyday computerized artifacts (Sellen, Rogers, Harper, & Rodden, 2009).

Transportation. Some of the most important artifacts in our lives are used to support human mobility. Cars, buses, trains, boats, planes, etc. have to be designed to enable their users to have effective, efficient, and safe transportation. Although some of these

artifacts are used by only one person, an analysis of dynamic transportation situations indicates that cognitive ergonomists must take a more global viewpoint that includes considering many persons who interact with each other through a large number of artifacts under conditions of severe time constraints and deadly hazards. Therefore, although ergonomics has been involved in the design of the cockpit of airplanes and cars (e.g., being recently involved in designing assistant systems for helping car drivers to drive safely), experts in this profession have also begun to be interested in the development of the whole sociotechnical system that has to be considered in air traffic or railway transportation. This is the area where most of specialized cognitive technical systems that support human activity have been developed and some of the crucial problems of interaction with artifacts were initially detected (see, in particular, the sections "Situation awareness and attention" and "Mental models," below).

Control processes. The area known as "process control" has a long tradition of research and practice in ergonomics. The term "process" refers here to that kind of industry known as "processing industry." A processing industry is one where energy and matter interact and transform one into another (Woods, O'Brien, & Hanes, 1987). A typical example of such an industry is nuclear power plants. But the paper production and the milk pasteurization industries also belong to this category. There is one ergonomically relevant characteristic that distinguishes among examples of processing industries. One can say that the various process industries differ in their degree of dependence on the artifacts that play a mediating role between the operators and the physical processes that they control. In the extreme case such as a nuclear power plant, all control must be done through artifacts. In many other cases, there may be a relatively direct relationship between human control operations (observations and/or actions) and the physical process.

Therefore, in general terms, in the process control domain one or more persons work to control a physical system using one or more artifacts. These individuals interact directly with the mediating artifacts, but not with the physical system that they are controlling. Unlike what happens in the interaction of a person with a computer when she or he is writing a text, in process control there is an external world, which is the physical industrial system that the person—or persons—perceive and control through the mediating artifact, which, of course, can be a computer (Ken'ichi, Kunihide, & Seiichi, 1997).

Intervention areas

Design. Design is at the heart of the ergonomist's profession (Dowell & Long, 1998). The design of a new system is the process that happens from the conceptualization of the artifact until when it is used by the people for whom it is intended. From the point of view of cognitive ergonomists, there are two aspects of interest in system design (Carroll, 1991). On the one hand, they are interested in "the process" of design itself. That is, cognitive ergonomists want to understand how people devise a new system, and what are the individual and group factors involved in making decisions that lead to certain solutions defining the system. Furthermore, cognitive

ergonomists would like to know whether the solutions adopted suit the needs and characteristics of users. Their main role in this sense is to describe the human being at all levels of functional organization appropriate for the system being designed (Velichkovsky, 2005; Wickens & Hollands, 2000). Therefore, cognitive ergonomists are interested in both the human being who designs and the human being interacting with the system that has been or has to be designed.

The work of cognitive ergonomists in the design process has undergone serious changes over the last decades. In the early days of human factor engineering, they were called on to explain why a particular design had not worked. Later on, they were called on to intervene directly in the design process (Wickens & Hollands, 2000). Today, the process of innovation requires that ergonomists “proactively” supply ideas and empirical data for the design of future artifacts improving human performance and public acceptance of new technologies (Akoumianakis & Stephanidis, 2003; Kohler, Pannasch, & Velichkovsky, 2008).

Technological innovation. The concept of “user-centered design” was developed during the 1980s in the design of technologies (Norman, 1986). Until then, system design issues were driven mainly by technological innovations. User-centered design aims at describing the human being who interacts with the system from the viewpoint of cognitive science. Then, based on those characteristics cognitive ergonomists provided engineers with a set of principles to be considered in the design. This paradigm has led to the establishment of usability research that has contributed greatly to the effectiveness, efficiency, and satisfaction of users in their interaction with the technologies and to a better interaction between users through technology (Holzinger, 2005).

However, one can at present witness a paradigm shift in the interaction that requires a change in how to understand and to evaluate usability. The change is motivated by the design of new applications and services under the influence of increasingly fast convergence of nano-, bio- and information technologies with cognitive science (see, e.g., *NBIC-Report*, 2006). This new development has several features that force cognitive ergonomists to rethink the concept of interaction. Those features are, first of all, that applications are designed to be used in intelligent and mobile environments. Secondly, they can be based on a wide spectrum of new materials, such as bidirectional organic glasses (transparent organic light emitting diodes, TOLEDs) and reactive polymers, making new forms of multimodal interaction with digital environments as well as ubiquitous computing possible. Last but not least, new applications, especially in time-demanding military and transportation domains, are supposed to be adaptable to the current intention, state of knowledge, and emotion of the user. The concept of “user-centred design” is becoming too non-specific as obviously the same user may be in different cognitive-affective states and may have changed intentions at different moments of time.

Safety and accident investigation. Historically, there have been three approaches to the study of human error. The first was taken from the field of engineering and led to the development of a range of techniques, generically called the “human reliability analysis” (HRA). These techniques are based on the assumption that the actions of

a person in a workplace can be considered from the same point of view as the operations of a machine. The objective is to predict the likelihood of human error and evaluate how the entire work system is degraded as a result of this error alone or in connection with the operation of the machines, the characteristics of the task, the system design, and the characteristics of individuals (Swain & Guttman, 1983). This approach has led to a considerable progress in the efforts to predict the occurrence of human error. However, it has been criticized as insufficient. Reason (1992) particularly notes that the main difficulty is the estimation of error probability. In designing new systems, there are no prior data on the error probabilities. One can count on data from simple components, such as errors that are committed reading a piece of data from a dial or entering it into a keyboard, but not the errors that may be committed by interacting with the system.

The second approach was adopted from cognitive psychology. In this, ergonomics seek to know the mental processes responsible for committing an error (Norman 1981; Reason, 1992). They assume that errors are not caused by irresponsible behavior or defective mental functioning. They may be rather the consequence of not having taken into account how a person perceives, attends, remembers, makes decisions, communicates, and acts in a particularly designed work system. This standpoint suggests investigating the causes of human errors by analyzing the characteristics of human information processing. Here, the first step has been the classification of errors according to the level of processing involved in the behavior that led to the error. Although there are more elaborate classifications today, it is possible to make a synthesis based on the classical scheme proposed by Jens Rasmussen (1983). He distinguishes three types of errors depending on the level and degree of cognitive control involved in the erroneous behavior. The three types of errors can be largely attributed to the familiarity that the person has with the system:

- *Errors based on skills:* When a person is very familiar with the task, her or his actions are overlearned as a low-level pre-programmed sequence of operations that do not require and often are not under conscious control. If one of these actions is poorly performed or the sequence is applied in an unusual order, a skill-based error occurs.
- *Errors based on rules:* The selection of actions in a situation often depends on the implementation of a set of rules of the type IF (condition) THEN (action). The activation of the right rules depends on the interpretation of the situational conditions. If a situation is misinterpreted the retrieved rule will be inappropriate as well. In other words, an error based on rules will occur.
- *Errors based on knowledge:* When we encounter a new problem situation, so that existing skills and learned rules are of little help, it is necessary to plan a novel action sequence to its resolution. This is a higher-order cognitive activity demanding a lot of conscious control. If the actions are not planned correctly, a knowledge-based error will occur.

Cognitive ergonomists took those classifications for granted in their attempts to find explanations of action slips and errors based on the cognitive models. However, the explanations have been criticized for being insufficient to facilitate the prediction

and prevention of errors. Indeed, it was difficult to predict errors from explanations based only on the hypotheses about human cognitive system. Therefore, a third approach has been developed recently to combine the reliability analysis developed by engineers with cognitive modelling. This approach starts from the basic assumption that the behavior of a person is determined by the context in which it occurs. The work system creates dynamic, ever-changing situations. It is therefore necessary to take into account the context and all levels of organization that contribute to system safety: the system's technology, the individual, the group, the organizational management, and cultural factors. In other words, it is not sufficient to estimate errors only from the perspective of human information processing (Wilpert, 2001).

According to this new approach, the person and her or his working environment should be considered as a highly interactive joint cognitive system (Hollnagel and Woods, 2007). The interaction between the two components is of a crucial importance for any ergonomic analysis. Based on these assumptions, several authors have proposed a methodology for estimating the probability of human errors depending on specific situation in which human-machine interaction occurs. The methodology presupposes two steps of analysis: (1) to identify the types of errors that are possible for a specific task in a given scenario of event development; (2) to classify these types of errors by their ranges of probability to identify which are the most probable and which are the least probable within the given joint cognitive system (Hollnagel, 1998; Cacciabue, 2004).

Theories and Models

In their everyday practical work ergonomists may well be more interested in improving what people do rather than what people know or feel. However an enduring improvement in performance seems to be possible only if the underlying cognitive representations as well as attitudes and competences of participating persons are known. This is why the Chomskian distinction between competence and performance becomes very important for cognitive ergonomists (Amalberti, 2001). In addition to this theoretical distinction, influential concepts are being borrowed, on the one hand, from ecological psychology and activity theory (Gibson, 1979; Leontiev, 1978) and, on the other hand, from the rapidly growing field of cognitive neuroscience (Hancock & Parasuraman, 2003).

Conceptual developments

With reference to Herbert Simon (1969), cognitive ergonomics sometimes is called “cognitive engineering.” Simon's book “The sciences of the artificial” had an enormous influence on the development of the discipline in the early 1970s. In this book and in a number of follow-up publications, he thoroughly argued that cognitive science must have its own area of application, i.e., the design of artifacts. To that end, he proposed the establishment of a “science of the artificial” related to cognitive science in the same sense as in the traditionally understood relationship between applied—engineering—disciplines and basic sciences.

Cognitive engineering deals with the problems of designing an effective mental work and the tools with which this work is done (Hollnagel & Woods, 1983). Therefore, the object of cognitive ergonomics is formulated around the concepts of “mental work” and the “cognitive tool (artifact).” Mental work is performed by cognitive systems that use knowledge to produce changes in the environment or domains of work. An artifact is any item manufactured by a human being with the purpose of improving any aspect of human behavior or mental operation. It can be a physical object (a computer mouse) or an abstract entity (a computer program) or an object representing a physical object (a drawing of a factory). The artifacts have been created to act on the environment, to modify any aspect of it, and to obtain information that allows us to know its characteristics and how it changed our actions.

Donald Norman (1986) was one who also argued for a combination of knowledge from cognitive science and engineering to solve design problems. According to him, the objectives of such a strategy would be twofold: (1) to understand the fundamental principles of human actions that are relevant to the development of principles of engineering design; and (2) to build systems that are pleasant to use. The first goal suggests a slight change in accents with respect to the original proposal of Simon. In fact, it put the discipline in line with the vision of some advanced experts in engineering (Vincenti, 1990): the establishment of cognitive engineering as a discipline of human action independent from, albeit related to, cognitive science from which it could borrow knowledge about cognitive processes. However, this proposal remained unattended for a decade, and ergonomics evolved according to Simon’s idea of understanding the cognitive engineering as an applied *pendant* to cognitive science. An example of this view can be found in some textbooks on human factors engineering (see, e.g., Wickens & Hollands, 2000), which are organized according to topics of human information processing. In this way, the list of chapters is the same as the list of chapters that can be found in any textbook of cognitive psychology.

A better acceptance of Norman’s ideas occurred when it became apparent how difficult it is to directly implement the results of cognitive psychology to explain the problems of design. These difficulties are currently causing a paradigm shift that will have repercussions beyond cognitive ergonomics itself, forcing a rethink of many fundamental issues that emerged from the cognitive revolution of the middle of the 20th century.

In the classical conceptualization, the artifact and the human being were considered independent from the context where the interaction between them took place. Thereafter, the aim of ergonomists was to study the characteristics of human information processing and provide designers with these data whenever they needed those characteristics to make decisions about the design of the artifact. This conceptualization has been called into question. First of all, due to technological progress, artifacts are designed today with a level of automation that enables them to be considered as (nearly) intelligent systems, almost at the same level as some (absentminded) human beings. Even if the last thesis can be disputed, contemporary artifacts often have a dynamic that is independent of the control actions that humans may impose on them. The difference, of course, is that artifacts are designed for human beings, while the human beings have evolved and been modified by evolution, development, and learning. Secondly, it is believed that human behavior is shaped by the sociotechnical

context in which it occurs and not only by the internal peculiarities of human information processing.

These considerations have been laid down in the paradigm of the joint cognitive systems (Dowell & Long, 1998; Hollnagel & Woods, 2007). The main message of the proponents of this approach is a broad interactionism: for a solution of cognitive design problems human behavior must be modelled as activity, in its interaction with the environment and with the other cognitive systems—both human and artificial—that there are in the environment. Therefore, in this new conceptualization of cognitive engineering the meaning of cognition itself is being reformulated in more dynamic and situational terms.

The meaning of cognition in cognitive ergonomics

In the traditional understanding, “cognition” refers to the acquisition, maintenance, and use of knowledge as examples of operations within the realm of human information processing. However, within dissident conceptions such as the joint cognitive paradigm, cognition should be understood in a broader sense, exceeding the limits of the individual’s brain or body. An example is the Gibsonian notion of “affordance,” which refers to all aspects of the environment supporting specific actions of individuals (Gibson, 1979). This notion is of obvious significance for cognitive engineering, to such a degree that some authors declare the design of affordances to be the main goal of human factors engineering (Vicente, 1999). In a similar vein, Norman (1986) stresses the importance of “external memory.” Under influence of these ideas, the meaning of cognition in cognitive ergonomics now refer to highly organized distributed systems (“distributed cognition”), such as the military, air traffic control, aircraft cabins, or navigation systems for large ships. Both people and artifacts are jointly regarded as agents within such a system. The focus is placed on the transfer and processing of information within and between agents. In this framework, cognition is viewed as a phenomenon that emerges from the work of the system as a whole (Hutchins, 1995).

One consequence of this redefinition has been the incorporation of theories that have been developed outside the mainstream cognitive research. This is the explanation for a discovery of activity theory (see Leontiev, 1978), which has its roots in European romanticism and Marxist philosophy. Activity theory, with its focus on the sociocultural origins of human thought and action, is now considered as a promising starting point for research in cognitive ergonomics (Nardi, 1996). Accordingly, there are no sharp distinctions between consciousness and behavior, and thus between external actions and internal thoughts, a distinction that is common for traditional cognitive science and ergonomics. Thoughts without external actions are considered as internalized social actions, similar to corresponding external actions (Vygotsky, 1978). As soon as the sociocultural context is considered, the scope of analysis becomes broader than in cognitive science. These contexts include communicating with others as well as the context in which the person is growing. The cultural context also includes the history of the artifacts, actions, and people.

The incorporation of new approaches and theories of cognition into ergonomics led to a discussion on the relative merits of macro- and microtheories whereby the

dominating view stressed the importance of the overarching explanations. For instance, Klein and colleagues (2003) proposed that cognitive ergonomists should create macrotheories that incorporate all the complexity of interaction within a socio-technical system. Simultaneously to this holistic trend, one can testify to the growing influence of concepts borrowed from the field of cognitive neuroscience. Being closely related to the progress in methods of brain and behavioral research, the second trend recently led to the development of neuroergonomics (Parasuraman & Wilson, 2008; Velichkovsky & Hansen, 1996). This tendency is especially evident, in the analysis of several traditional topics of human factors studies which are discussed below.

Conceptual topics

Situation awareness and attention. In performing complex tasks such as flying an airplane, it is necessary to process huge amounts of data on what is happening in the cabin, where there may be more members of the crew, in the air outside the plane, and even at the airport and the control tower. This information must be attentively handled, retained, interpreted, and used to make necessary decisions for the airplane to fly properly and land safely. The processes of conscious perception and understanding of the situation are generically called “situation awareness.” In many application domains, ergonomists need this concept to describe cognitive processes that are responsible for the acquisition, storage, and use of information available to the person doing the work. Although there are several ways to define what situation awareness is, the model proposed by Endsley (1995) is widely accepted today. According to this author, situation awareness is the perception of elements in the environment within a volume of time and space, the comprehension of their meaning and projection of their status in the near future. Therefore, it includes three levels of information processing: (1) perception of the elements of the environment; (2) understanding the current situation; and (3) predicting (projecting) the situation’s future development.

One of the reasons for this rapidly growing interest in situation awareness is the instability of human performance related to the automation of work processes. The problems—or “ironies”—of automation were first noted by Lisanne Bainbridge in a seminal paper (Bainbridge, 1983). With a high degree of automation, the human operator is out of loop of controlling processes. As a result, operators are less well practiced in their abilities to take over the process when an automatic unit fails. This deterioration results from the fact that the manual and cognitive skills decline due to the absence of active participation in the process. Furthermore, it becomes more difficult with progressing automation to gain access to knowledge about system behavior. Many authors see the solution to such problems in adaptive automation, which could take the current state of knowledge of human operator into account and, in this way, support a better division of labor between humans and machines. However, the solution presupposes reliable and timely feedback information about human understanding of the situation. This is the area where neuroergonomics seems to have serious chances of success (Parasuraman & Wilson, 2008). In particular, neurocognitive studies of attention build the main source of knowledge about

mechanisms of situation awareness. These studies have elucidated three different attentional networks in the human brain (Posner, Rueda, & Kanske, 2007) and up to six levels of cognitive organization (Velichkovsky, 2005). Changing the balance of these networks can explain fluctuation in the level of human performance over time, as in the case of driver's behavior in hazardous situations (see Velichkovsky, Rothert, Kopf, Dornhoefer, & Joos 2002).

There seems to be a new understanding in ergonomics that some degree of attention and situation awareness is always required to control the performance of any task, no matter how seemingly simple and safe it is. Today this is a topic of vital importance in many areas of ergonomics from military applications, industry, and transportation to the work of medical professionals. For example, a recent WHO-funded study has shown that the rate of postoperative mortality in a number of hospitals across the world could be reduced by nearly 40% if before the surgery medical personnel had answered questions from a simple checklist of situation awareness: "Are we at the right place here?", "Is that our patient?", "What organ [are] we going to operate on?" (Haynes et al., 2009).

Mental models. When interacting with a system, people normally have some knowledge of its structure and functioning. This small-scale subjective representation of system's structure and functioning is called a mental model (see Johnson-Laird, 1983). Taking into account the peculiarities of users' mental models in the design of artifacts is considered to be crucial for an efficient interaction. Therefore, the investigation of mental models is one of the central themes in cognitive ergonomics (Cañas, Antolí, & Quesada, 2001; Ken'ichi et al., 1997).

Mental models have been studied in applied psychology and ergonomics from several different perspectives. First of all, it is normally expected that having a relatively exact mental model facilitates the acquisition of manual skills for dealing with a physical system. More recently, interest has been extended to the supervision of automatically controlled systems, where the skills that come into play are cognitive operations of detection, diagnosis, and compensation for failures that the system may have. Computational analysis of mental models complexity has been widely used to predict the understanding of instructions that describe how to deal with a technical system. Finally, in the area of HCI, researchers have consistently proven that when a person interacts with the computer she or he acquires knowledge about its structure and operation. Interestingly, this acquisition may be less efficient with relatively easy-to-use graphical user interfaces than with old-fashioned command line interfaces. Other research has shown that the acquisition of an adequate mental model of the computer facilitates learning a programming language (Cañas, Bajo, & Gonzalvo, 1994; Kieras & Bovair, 1984; Navarro & Cañas, 2001).

Learning with and about artifacts. The topic of learning is so important in ergonomics that it cannot be ignored in any attempt to establish a conceptual framework for this discipline. However, one has to distinguish between several related subtopics and address them separately. First of all, there is the basic issue of the variety of learning. As in the case of attention, discussed above, there seem to be at least three neurocognitive systems participating in different forms of skill and knowledge acquisi-

tion. Norman (1986) describes three forms of learning in the following terms: (1) *accretion*, that is, an accumulation of factual knowledge such as usually takes place during classical academic education; (2) *tuning*, a gradual adaptation in the parameters of our actions, leading to a steady—albeit not necessarily monotonic—improvement in performance; and (3) *restructuring*, which is a form of higher-order learning by discovery of new conceptual relationships. It should be noted that the last form of learning is directly related to the issue of conceptual change that is of paramount importance both in science education and in learning to work with new technological tools (Chi & Roscoe, 2002).

On a more practical level, after designing a new artifact users must learn to interact with it. This fact has become a crucial issue in modern industry due to two factors. First, new artifacts are increasingly more complex requiring the development of more complex cognitive abilities. Second, technological progress is so fast that nobody can expect to learn to interact with an artifact and continue using it for long time. For example, a couple of decades ago we were writing with typewriters, and now we do that with computers and text editing programs that have changed during these years in a radical way. Thus, in a conceptual scheme of the joint cognitive system, the topic of learning is as important as the topic of design itself. In this sense, much work of cognitive ergonomists has been and will remain focused on the acquisition of the mental models of artifacts, which we discussed previously.

Another major issue of learning is the designing of teaching aids. The people responsible for education have always used educational tools—from the blackboard and chalk to modern “intelligent tutoring systems”—to do their work (Scaife & Rogers, 2005). Educational institutions and companies have followed a classic approach until recently, that is, training in classes with students and teachers in the same space at the same time, with face-to-face communication. In the global economy, computer-mediated communication and distributed work environments are becoming increasingly important. In particular, large enterprises maintain widespread operations and large numbers of employees working on the same project and with similar tools, although they are geographically separated. This has now made feasible the programs of distance learning that are based on the computer imitation of a presence in the same spatial environment (as virtual or mixed reality) as an important option (Kohler, Pannasch, & Velichkovsky, 2008). Simultaneously, universities are trying to find proper technological tools of bringing educational training to all places where there might be a student or an employee who requires it (Salas & Cannon-Bowers, 2001).

Decision making. One of the most active areas of human factors engineering from the beginning has been what is known as the “control of critical incidents.” This category includes subtopics such as fire control, medical emergencies, control centers and civil emergencies, military operations, stock market crises, the rescue of victims of kidnappings, and so forth. In dealing with these issues, ergonomists have been using several terms that could be considered at least partial synonyms: “command and control,” “dynamic decision-making,” “distributed decision-making,” “natural decision-making,” and “decision science” (Artman, 1998; Brehmer, 1992; Zsombok & Klein, 1997). In all these cases there is a common feature, which is that the persons

who have to control these incidents should make decisions under time pressure and with little information to prevent (often) catastrophic consequences. Therefore, although control of critical incidents involves a number of complex psychological processes, decision making is considered to be the most important of them.

Several interrelated trends are characteristic for contemporary decision-making studies in ergonomics. Mainly, their authors turned away from tasks borrowed from formal logics to more naturalistic situations. As noted by Beach and Lipshitz (1993), psychologists have been traditionally focused on the endpoint of decision making: the choice between alternatives that are already given by the task conditions. They have usually investigated inexperienced subjects and assessed the quality of decisions according to “rational” criteria from context-free models. There are reasons to doubt the applicability of results found in these studies to real-world situations (Hutchins, 1995). Indeed, if the conditions are changing, as in the control of critical incidents, the findings may be of marginal value. In natural situations, people are highly experienced and have to generate new alternatives under time pressure. Moreover, the training of professionals, which is based on formal algorithms of decision making, can be rather misleading as the need to take a quick and obvious decision leaves no time to contrast it with other theoretically possible moves (Salas, Cannon-Bowers, & Johnston, 1997). The combination of time pressure and the highly significant outcomes explains the interest that decision science demonstrates in “hot,” (i.e., affectively loaded), rather than to “cold” cognition (see e.g. Kahneman, 2003).

Organizational processes. Decision making in a group and its relations with the leadership and related matters (see, e.g., Heller, 1992) is the central topic of organizational psychology, which is discussed in one of the chapters of this handbook. Contemporary ergonomics has taken a slightly different stance, by concentrating efforts on analysis and optimization of the role that knowledge has on the productive activities of human groups. To this end, the term “knowledge management,” which refers to the process of generation, storage, and retrieval of knowledge within a social-technical system, is often used. This topic is of increasing importance for cognitive ergonomics since knowledge management requires the use of high-tech artifacts. While it is true that large machinery began to be manufactured in the industrial revolution in the 19th century, and products that were manufactured by human beings were physical objects, today everybody recognizes that knowledge is also generated by human activity. Recognition of this fact is now so general that it is easy to hear that a lot of socioeconomic activity aims to produce knowledge. Furthermore, an obvious fact is also recognized: in the production of objects, organizations generate knowledge of these objects and of the processes of manufacturing and distribution.

The role that knowledge plays can be analyzed from many perspectives, but from the point of view of cognitive ergonomics one has to consider three of them. Firstly, there are many human activities where knowledge is explicitly generated in the development of products. Secondly, there is valuable knowledge that is generated implicitly as people are continuously involved in their activities. This is true of most productive activities at all levels and in all sectors. For example, in a factory where there are certain objects with which workers gain personal experience in a production process, that experience is not acquired in the training courses given by the company, or in

the manuals of the machines' producers. That knowledge is often lost when a worker is removed from a job due to retirement, illness, or relocation. Finally, there are countless activities where knowledge must be shared among members of a real or virtual organization. This is the case, for example, in a civil emergency in which members of a professional group must put their knowledge together, communicate, and coordinate their actions to achieve a common goal in order to provide the service required by society.

Mental workload and stress. All work requires human effort. Whenever a person is faced with a task, she or he must implement cognitive resources leading to a variety of functional states that are known in ergonomics by the terms "mental workload," "fatigue," "stress," and the like. Ergonomists have always had an interest in this topic since it was recognized that every time a person has to perform some task that requires more resources than those that she or he has available, that person will be overloaded, and that can affect work performance and, in a long run, her or his mental and physical health. Even if the development of specialized human-computer interfaces has been intended to reduce the demands for cognitive resources, sometimes tools developed to assist people impose additional burdens on them leading to excessive stress and loss of situation awareness with disastrous consequences. There are, of course, ample cases of bitter ironies known from the history of many half-automated systems in transportation and elsewhere.

The concept of mental workload was proposed in the 1970s to explain how cognitive resources are allocated and coordinated to perform several tasks simultaneously. After all these years, the concept has proven to be an important but somewhat vague instrument in the work of ergonomists. It would be important to have an exact and measurable definition of human cognitive limitations during engineering of new systems allowing designers to predict which implementation will maximize the effectiveness and still leave the user a residual capacity to cope with unexpected demands (Yeh & Wickens, 1988). In addition, the labor legislation of industrialized countries recognizes that mental workload affects mental and physical health. Therefore, the law requires companies to evaluate the mental workload to which workers and employers are exposed.

One finds references to mental load in the research areas that have traditionally been identified as "dual task performance," "task execution time-sharing," "task switching," and "working memory." The last notion is used because it combines the idea of a fixed amount of cognitive resources with active operations on them. For example, it has been argued that many of the mistakes made when interacting with a computer are caused by an excessive load of working memory (Olson & Olson, 1990; Gevins et al., 1998). The field of study has been recently extended to increasingly complex multitasking situations that share characteristics with dual tasks, but also possess other features that require high-level coordination during the information processing.

However, there are some persistent problems with the notion of mental workload. First, this is an overtly mentalist concept. Second, the nature of cognitive resources and their relations to structural and operational constraints of processing remain unknown. Numerous hypotheses, models, and theories have been proposed to clarify

these issues (Meyer & Kieras, 1997). One of the widely accepted ones is the model of multiple resources by Christopher Wickens (1984). According to it, there is more than one kind of resource, such as those enabling verbal and nonverbal processing. If tasks share the same resources and are performed concurrently, increasing the difficulty of one of them will lead to decline in their performance. If each task addresses different pools of resources, manipulation of the difficulty of one of them would not affect performance. Thus, the combination of resources within and between cognitive dimensions determines the level of mental load. However, the exact nomenclature of cognitive resources is also unclear. The hopes for progress in understanding limitations of human information processing are currently related to functional brain imaging studies (Hancock & Parasuraman, 2003).

Any limitation in performance should also be considered from the perspective of a variety of human functional states such as fatigue, monotony, and stress (Leonova, 1998). Stress is increasingly the major concern in the life of industrialized nations. Time shortage, available skills, and the subjective significance of the task are the main factors causing subjective feelings of acute stress. Both the conceptual basis of contemporary stress research and its methodology—ranging from subjective questionnaires to biochemical analysis of “stress hormones” and genomic studies—seem to demonstrate a positive contrast to many other areas of empirical investigations in ergonomics and human factors engineering. Within this interdisciplinary context, a consistent interest in individual differences opens the way to an understanding of the mechanisms of human stress resilience and its prediction in borderline conditions.

Methodologies

The methods used in ergonomics aim to explain and predict the consequences of taking certain decisions during the design of a sociotechnical system (i.e., the introduction of a new artifact). In this sense, the methodology does not differ in general terms from those used in cognitive psychology and other human factors disciplines such as anthropology and sociology. However, it is necessary to clarify that for ergonomists any explanation has to be followed by prediction. An important aspect of ergonomic methods is that they can be applied before or after the introduction of a change in the system. When they are applied before the introduction of the change, the aim is to explain and predict the consequences of change. When they are applied after the change, the goal is usually to explain and evaluate those consequences in terms of the effectiveness of this change in improving the interaction between system components.

Ethnographic method and field studies

The ethnographic method is applied when ergonomists have to analyze a completely unknown situation (Garfinkel, 1967). Being initially used in anthropology and sociology, it represents a kind of immersion of the researcher in the environment to describe and explain the observed phenomena. The emphasis is on observation relatively free of theory and on a “qualitative” rather than “quantitative” description of what is

observed. This approach emerged to replace methods based on structural interviews and questionnaires. It is argued that interviews and questionnaires lose an important aspect of the task, which is the implicit knowledge closely tied to the routines, trials, and intuitive personal adaptations. Since this knowledge is unconscious and is behind rapid responses in real tasks, people are simply too overloaded to make it explicit when answering questionnaires and interviews. In addition, questionnaires and interviews seem to be not truly helpful when trying to analyze the interrelationships between several persons.

Researchers using the “ethnomethodology” often argue that their observations are not driven by any assumptions. However, it is difficult to believe that ergonomists would be able to shed all their prior knowledge to observe a situation without bias (Shapiro, 1994). Without going into details of a methodological discussion that is raging in the philosophy of science, many ergonomists prefer to use a well-established method, called field study, in which as in the ethnographic method, one observes and describes a situation without seriously interfering with it, but the observations are guided by assumptions that are explicitly established from the outset.

Standards and evaluation testing

In many cases, a task can be easily categorized as belonging to one or another standard situation. For example, when introducing a new software application within the common platform of a graphical user interface, one has to consult the corresponding guidelines on designing dialogs and the overall requirements for human–computer interaction (Karwowski, 2005). The International Standardization Organization (ISO) is the highest instance in the hierarchy of industrial, national, and international bodies involved in regulation of human interaction and work with artifacts. In specific situations, one should consult standards developed by other institutions, such as the Commission Internationale de l'Éclairage (CIE) which is the authority on light, illumination, color, and color spaces.

Even if the application of standards is not immediately possible, the theoretical and empirical development of human factors and ergonomics allows analysis of artifacts by use of known principles and data without carrying out experimental research. For example, there is sufficient documented knowledge about human sensory and motor systems to make it unnecessary to conduct a new experiment every time there is a need to design a new display or a mouse (Boff, 1986). In addition, there are a large number of reference sources that respond to the growing need for specific instruments and methods for testing the usability of human–system interfaces (Charlton & O'Brien, 2002; Holzinger, 2005). All indications are that the emerging market for information on these tools and standardized techniques will continue to grow at a fast pace in the 21st century.

Experiment

When a system is being designed or evaluated, it is possible to propose several design alternatives. These alternatives can be considered experimental hypotheses and tested

by conducting a controlled experiment as is usually done in psychology and in most of natural sciences. In a multivariable laboratory experiment, one can easily change the color and the number of the menu items in a word processor to see their main effects and their interaction on the time that a group of users needs to select an option. The big problem with the experimental method is a trade-off relationship between ecological validity and necessary control of independent variables. As a high degree of ecological validity is of primary importance for ergonomic research, ergonomists often have difficulties in applying experimental methods. For example, when designing a new interface to be used by air traffic controllers, ergonomists are faced with the following characteristics that make it difficult to run experiments: (1) The subjects are experts in implementing the tasks to be analyzed and they depend on their existing skill and knowledge; (2) The tasks are complex and cannot be divided into linearly ordered components; (3) It is impossible to perform a long series of similar trials and preserve the subject's state of attention and motivation; (4) The behavior is complex and cannot be decomposed, so that simple imposed answers do not interfere with higher-order, cognitive components of behavior; (5) Implementation of the tasks to be studied depends on the overall situation that cannot always be repeated; and (7) The tasks are distributed among several people, each one with her or his own role, so that the complexity of experimental design and data analysis is additionally multiplied.

Still, there seems to be no method—in the immediate and middle-term perspective—that could better fulfil the task of investigating scientifically based human factors and ergonomic research than experiment. Paradoxically, one can achieve a higher applicability of experimental data by more in-depth laboratory research. For example, the promises of neuroergonomics are related to establishing the brain mechanisms of different forms of attention (Posner, Rueda, & Kanske, 2007). The success of direct diagnostics of their state of activation and the emerging techniques of brain-computer interfaces depends on the progress made in cognitive science. In the longer run, one can hope to replace most real experimental work by running computational experiments with virtual artifacts and users.

Simulation

Simulation of the situation. One way to conduct experimental research and to avoid the problem of ecological validity is by simulating real-world situations. This is what researchers are currently doing in complex situations such as air traffic control and nuclear plant control. Although simulators are used primarily to train people who work in these situations they can also be used to perform experiments with them. Simulators are a powerful tool for experimental research of complex behaviors in the laboratory. With the development of computer graphics and the technology of virtual reality, it is possible to simulate real situations with sufficient psychological validity. Furthermore, since the computers themselves are becoming increasingly present in the actual situation, to work with them is relatively easy. For example, in the control room of a nuclear power plant, controllers work primarily with computers operating keyboards, mouse, and so on. Therefore, to build a simulator of a control room to study these processes, it is only necessary to create computer

programs that simulate the events that occur in the plant. The fundamental problem with simulators is that it is not easy to reproduce the emotional stress of real situations.

One can distinguish three types of simulators that can be used in experimental research: entertainment simulators (video games), training simulators, and those explicitly designed for experimental research. The last are often the most appropriate because they have components that the researchers can use to change parameters of the situation that are of interest to them. These simulators are also designed to reflect how accurate the responses of subjects are and to measure other dependent variables such as reaction time.

Simulation of the human being. Ergonomists study complex behaviors that are difficult to dissect (Klein et al., 2003). They are also interested in a broad range of phenomena to predict human behavior and functional states under sometimes hazardous conditions. In addition, many industrial artifacts nowadays are first produced in a fast-prototype manner, as a virtual mock-up suitable for some forms of usability testing. An ideal counterpart of this partially virtual world would be, of course, a virtual human dummy that implements some of the essential characteristics of potential users. The contemporary efforts along these lines concentrate on the biomechanical and optical features of human beings (Duffy, 2008).

It is obviously more difficult to create computer simulations of the human mind, so that cognitive ergonomists could make predictions about other forms of observable behavior only on the basis of computational experiments. There are two main areas of methodical progress in view of the simulation of cognitive processes and representations. The first area is that of methods of psychosemantics and knowledge engineering (Kendal & Green, 2007). They are used to elucidate semantic representations an individual or a group have about some domain of activity. The methods can be applied as an early step in building expert systems. Work done within the second area envisages the modeling of human cognitive processing as a whole. These computational simulations usually belong to one of the following global cognitive architectures: ACT-R (Anderson et al., 2004), SOAR (Newell, 1990), CCT (Kieras & Polson, 1985) and EPIC (Meyer & Kieras, 1997). A relatively recent requirement for these models is that they are also involved in neurocognitive testing, delineating a road to a neuromorphic modeling of the user's mind.

Conclusion and Future Issues

Today ergonomics is an established discipline where professionals with interdisciplinary backgrounds work together in designing sociotechnical systems with the common goal of fitting them to human needs and well-being. The work is embedded in the activities of international scientific associations such as the IEA, HFES, and EACE as well as the International Association of Applied Psychology (IAAP). The reports on ergonomic research and applications can be found in a number of well-cited journals—*Ergonomics*, *Human Factors*, *Le Travail Humain*, *Human Technology*,

and *Cognition, Technology and Work*, to name but a few. Most of the leading universities around the world offer undergraduate and graduate programs in which students are trained to become human factors and ergonomics professionals. The Professional Standards and Education Committee of the IEA certifies ergonomists, maintains a directory of ergonomics educational programs and provides guidance on the quality of professional education.

The future agenda of human factors research and applications is set up by how technology will be developed and used in society. At the high end of technological development, there will be many options to meet the human factors challenges as they were carefully listed on eve of the new millennium (Nickerson, 1992). It will inevitably come to a further convergence of the basic technologies with the resulting enhancement of human performance (*NBIC Report*, 2006). One can expect that usability evaluation will be soon evolved to a more scientifically based and predictive endeavor. Another expectation is that of a proliferation of completely new forms of interface. Some of them may have nano-dimensions fulfilling their roles within the molecular machinery of human cognitive-affective processes. With a high probability, neuroergonomics will not long have the status of the youngest “science of the artificial” perhaps being combined with something like computational ergonomics. Due to the uneven pace of these processes, there will, however, be regions and domains, where people will still have to perform hard, dirty, unpleasant physical tasks around the clock without proper gratification. Here human factors experts and ergonomists, along with politicians and social workers, should seek to improve the work environment of such individuals by more traditional means (Hancock & Parasuraman, 2003).

All these processes, whether they are directly responding to people’s needs or are driven by some inherent logic of development in science and technology, have a tremendous influence on our life. Their future influence on the cultural and political transformation of human society will be comparable with the impact the introduction of the internet and mobile communication have had during the last two decades. By continuing their service to society, experts in human factors and ergonomics will obviously have a particularly attentive look at the human side of our interaction with machines, computers, and other increasingly sophisticated artifacts. It remains to be hoped that their professional efforts will lead to a better fulfilment of human needs and values in the years to come.

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